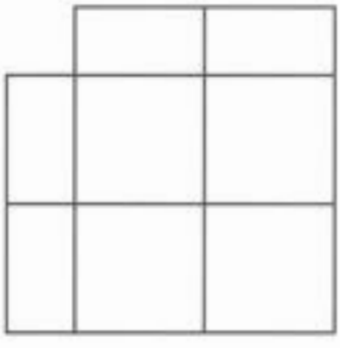
Sex and Genes

42A



Norman human body (somatic) cells contain 23 pairs of chromosomes. In 22 of the pairs, both chromosomes look the same as each other. In one pair, the chromosomes do not always resemble each other. In females, these chromosomes are the same, and are called X chromosomes. In males, one of the chromosomes looks the same as the X chromosome in the female cells. The other looks different and is called the Y chromosome. So, human females have 22 pairs of 'normal' chromosomes and an XX pair. Human males have 22 pairs of chromosomes and an XY pair. Extra sex chromosomes are found in about one in 1000 people. The three most common examples of this are XXY, XYY and XXX.

1 Complete the Punnett square on the right to show what the ratios of sex (gender) genotypes in offspring will be between a human male and a human female.



2 From your completed Punnett square, what is the ratio of males to females?

_____

3 How many pairs of chromosomes do humans have in somatic cells?

_____

Genes are grouped together on chromosomes. Those found on X or Y chromosomes are said to be 'sex-linked'. The Y chromosome doesn't carry very many genes. It does carry the information to give male characteristics.

4 Name two male secondary sexual characteristics. _____

In males, nearly all the genes on the X chromosome, whether or not they are dominant, will affect many functions, not necessarily related to sex. This means that there is only one form of most genes on the sex chromosomes of males.

Characteristics controlled by genes on the X chromosome are sex-linked because they occur more often in one sex than the other.

Another way of saying this is that the XX chromosomes in a female contain many paired genes. The Y chromosome in a male does not have partners for most of the genes on the X. So any recessive genes on the X chromosome will occur more often in males. These unpaired genes on the X chromosome, are sex-linked.

5 What is a *recessive* gene? _____

_____

If there is a defective gene on a part of the X chromosome of an XY pair, there sometimes isn't a corresponding gene on the Y portion. There is no 'backup' (such as a dominant allele) and symptoms will result. Over 50 inherited diseases are sex-linked. Examples of these are Duchenne Muscular Dystrophy (DMD), red-green colourblindness and haemophilia (a disease in which blood doesn't clot). The disease DMD is caused by a break in the X chromosome at the site of the DMD gene. Over 80% of haemophilia is family-linked, while other cases can be caused by spontaneous mutations.

A gene related to colour vision is found on the X chromosome, but not on the Y chromosome. This is a recessive gene.

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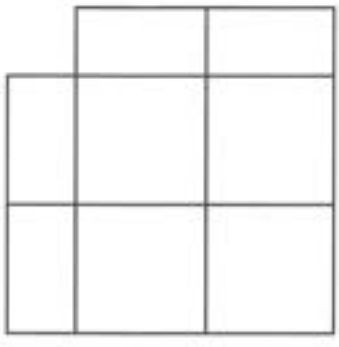
Sex and Genes

42B

Look at the possible offspring resulting from a mother who carries one gene for colour blindness (and will not be colour blind herself) and a 'normal' father who doesn't carry any genes for the disease.

If we represent the colour blindness gene as X^c and the normal gene as X, the mother is X^cX and the father is XY.

6 Complete the Punnett square on the right to show the genotypes of the offspring of the X^cX mother and the XY father.



7 From the Punnett square you completed, what are the proportions of the phenotypes of the offspring? State whether these are male or female.

_____

_____

8 What do we mean by a 'carrier' for a genetic disease?

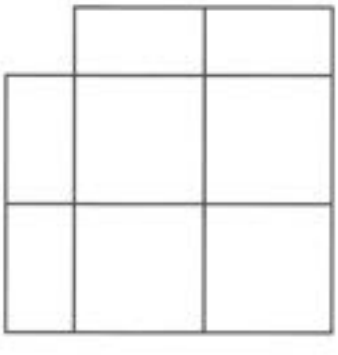
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Drosophila melanogaster (a type of fruit fly) are small two-winged flies also known as 'red-eyed pomace flies'. They are often used in genetics experiments because they breed quickly. Thomas Hunt Morgan (1866–1945) was an American geneticist, who carried out studies of heredity in *Drosophila* and was awarded a Nobel Prize in 1933 for discovering that genes for specific traits were located at specific sites on chromosomes. He experimented with eye colour.

He found that eye colour (red or white) is a sex-linked inheritance in *Drosophila*. Red eyes are dominant (R) over white eyes (r).

9 Using the Punnett square on the right, work out the possible genotypes of a cross between a white-eyed female (X^rX^r) and a red-eyed male (X^RY).



10 From the Punnett square, what are the proportions of the phenotypes in the offspring? State whether these are male or female.

_____

_____

11 Name an inherited disease, apart from colour blindness, that is sex linked._____

12 Why do geneticists use fruit flies to carry out experiments?

_____

13 What is the common name for *Drosophila melanogaster*? _____

14 Explain the difference between the way that inherited diseases that are 'sex-linked' are passed on and those that are due to recessive genes are passed on.

_____

_____

15 What are the symptoms of haemophilia?_____

16 'About one in a thousand people will have three sex chromosomes.' True or false?_____

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